

Ibn Tofail University

Algebra IV

Problem Set III

Exercise 1:

Let $A = \begin{pmatrix} -2 & 2 & -1 \\ -1 & 1 & -1 \\ -1 & 2 & -2 \end{pmatrix} \in M_3(\mathbb{R})$ and $B = \begin{pmatrix} -1 & 1 & 1 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$.

1. Determine the characteristic polynomial of A , then a basis of each sub-space.
2. Give the Jordan reduction J of A , the minimal polynomial of A , and the invertible matrix P such that : $A = PJP^{-1}$
3. Calculate J^n and deduce A^n .
4. Prove that A and B are the same.

Correction

1. Characteristic polynomial:

$$\chi_A(\lambda) = \det(A - \lambda I_3) = \det \begin{pmatrix} -2-\lambda & 2 & -1 \\ -1 & 1-\lambda & -1 \\ -1 & 2 & -2-\lambda \end{pmatrix} = -(\lambda + 1)^3$$

Eigenvalue: $\lambda = -1$ (triple).

Kernel of $A + I_3$:

$$A + I_3 = \begin{pmatrix} -1 & 2 & -1 \\ -1 & 2 & -1 \\ -1 & 2 & -1 \end{pmatrix}$$

All rows are identical, so $\dim(\ker(A + I_3)) = 1$. Basis: $\left\{ \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \right\}$.

Kernel of $(A + I_3)^2$:

$$(A + I_3)^2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

So $\dim(\ker((A + I_3)^2)) = 3$. We choose basis of Jordan chain:

$$v_1 \in \ker(A + I_3)^2 \setminus \ker(A + I_3), \quad v_2 = (A + I_3)v_1 \in \ker(A + I_3)$$

Let $v_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, then $v_2 = (A + I_3)v_1 = \begin{pmatrix} -1 \\ -1 \\ -1 \end{pmatrix}$. Then take $v_3 = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$ (from $\ker(A + I_3)$ and linearly independent from v_2).

2. Jordan form:

$$J = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad P = \begin{pmatrix} 1 & -1 & 2 \\ 0 & -1 & 1 \\ 0 & -1 & 0 \end{pmatrix}$$

Minimal polynomial: $m_A(X) = (X + 1)^2$.

3. Powers:

$$J^n = \begin{pmatrix} (-1)^n & n(-1)^{n-1} & 0 \\ 0 & (-1)^n & 0 \\ 0 & 0 & (-1)^n \end{pmatrix}$$

Then $A^n = PJ^nP^{-1}$.

4. Similarity: Both matrices have same characteristic polynomial $-(\lambda + 1)^3$ and same minimal polynomial $(\lambda + 1)^2$, and same Jordan form, so they are similar.

Exercise 2:

Let

$$A = \begin{pmatrix} -4 & 1 & 0 & 1 \\ -2 & -1 & 0 & 1 \\ -12 & 6 & 3 & 1 \\ -2 & 1 & 0 & -1 \end{pmatrix}$$

1. Determine the eigenvalues of A .
2. Determine the dimensions of the sub space associated to the eigenvalues of A .
3. find the Jordan reduction matrecice of A .
4. what is the minimal polynomial of A .

Correction

1. Characteristic polynomial:

$$\chi_A(\lambda) = \det \begin{pmatrix} -4 - \lambda & 1 & 0 & 1 \\ -2 & -1 - \lambda & 0 & 1 \\ -12 & 6 & 3 - \lambda & 1 \\ -2 & 1 & 0 & -1 - \lambda \end{pmatrix} = (\lambda - 1)^2(\lambda - 3)^2$$

Eigenvalues: $\lambda_1 = 1$ (double), $\lambda_2 = 3$ (double).

2. Dimensions: For $\lambda = 1$:

$$A - I_4 = \begin{pmatrix} -5 & 1 & 0 & 1 \\ -2 & -2 & 0 & 1 \\ -12 & 6 & 2 & 1 \\ -2 & 1 & 0 & -2 \end{pmatrix}$$

Rank = 2, so $\dim(\ker(A - I_4)) = 2$. Already diagonalizable for this eigenvalue.

For $\lambda = 3$:

$$A - 3I_4 = \begin{pmatrix} -7 & 1 & 0 & 1 \\ -2 & -4 & 0 & 1 \\ -12 & 6 & 0 & 1 \\ -2 & 1 & 0 & -4 \end{pmatrix}$$

Rank = 3, so $\dim(\ker(A - 3I_4)) = 1$. Need Jordan block of size 2.

3. Jordan form:

$$J = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 3 \end{pmatrix}$$

4. Minimal polynomial: Since $\lambda = 1$ gives 2D eigenspace and $\lambda = 3$ gives Jordan block of size 2:

$$m_A(X) = (X - 1)(X - 3)^2$$

Exercise 3:

Let $A \in M_8(\mathbb{C})$ and it's minimal polynomial is $X^2(X - 1)^3$ and the characteristic polynomial $X^4(X - 1)^4$. Let J be the Jordan reduction of A .

What's the possible formes of J ?

Correction

Given: - Characteristic polynomial: $X^4(X - 1)^4 \rightarrow$ eigenvalue 0 with algebraic multiplicity 4, eigenvalue 1 with algebraic multiplicity 4. - Minimal polynomial: $X^2(X - 1)^3 \rightarrow$ largest Jordan block for eigenvalue 0 has size 2, for eigenvalue 1 has size 3.

Possible Jordan forms:

For eigenvalue 0 (multiplicity 4, largest block size 2):

- Possibility 1: Blocks of sizes 2, 2
- Possibility 2: Blocks of sizes 2, 1, 1

For eigenvalue 1 (multiplicity 4, largest block size 3):

- Possibility 1: Blocks of sizes 3, 1
- Possibility 2: Blocks of sizes 3, 1, ... but total size 4, so only 3,1 possible
- Actually also possible: 2,2 or 2,1,1 but largest must be size 3, so only 3,1 works

Wait, largest block for eigenvalue 1 is size 3, so possible partitions of 4 with max 3:

- 3,1
- 2,2 (largest block would be 2, not 3)

- 2,1,1 (largest block would be 2, not 3)
- So only 3,1 possible for eigenvalue 1

Thus the possible Jordan forms are:

$$J = J_0 \oplus J_1$$

where J_0 is either:

- $\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$ (blocks 2,2)
- $\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$ (blocks 2,1,1)

and J_1 is always:

$$\begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (\text{blocks } 3,1)$$

So two possible Jordan forms in total.

Exercise 4:

Let

$$A = \begin{pmatrix} 1 & \alpha & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 2 & 3 & 1 \\ -2 & -(4 + \alpha) & -1 & -2 \end{pmatrix}$$

where $\alpha \in \mathbb{R}$.

Give J and P Based on α . ($A = PJP^{-1}$).

Correction

Characteristic polynomial:

$$\det(A - \lambda I) = \det \begin{pmatrix} 1 - \lambda & \alpha & 0 & 0 \\ 0 & 1 - \lambda & 0 & 0 \\ 1 & 2 & 3 - \lambda & 1 \\ -2 & -(4 + \alpha) & -1 & -2 - \lambda \end{pmatrix} = (1 - \lambda)^2(\lambda - 2)(\lambda + 1)$$

Eigenvalues: $\lambda = 1$ (double), $\lambda = 2$, $\lambda = -1$.

For $\lambda = 1$:

$$A - I = \begin{pmatrix} 0 & \alpha & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 2 & 2 & 1 \\ -2 & -(4 + \alpha) & -1 & -3 \end{pmatrix}$$

Case 1: $\alpha = 0$ - Then $\text{rank}(A - I) = 2$, so $\dim(\ker(A - I)) = 2 \rightarrow$ diagonalizable for eigenvalue 1.

Case 2: $\alpha \neq 0$ - Then $\text{rank}(A - I) = 3$, so $\dim(\ker(A - I)) = 1 \rightarrow$ Jordan block of size 2 for eigenvalue 1.

Jordan form:

$$\text{- If } \alpha = 0: J = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

$$\text{- If } \alpha \neq 0: J = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

Matrix P : We find eigenvectors and generalized eigenvectors:

$$\text{For } \lambda = 2: \text{ Solve } (A - 2I)v = 0, \text{ find } v_1 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix}$$

$$\text{For } \lambda = -1: \text{ Solve } (A + I)v = 0, \text{ find } v_2 = \begin{pmatrix} 0 \\ 0 \\ -1 \\ 2 \end{pmatrix}$$

$$\text{For } \lambda = 1: \text{ - If } \alpha = 0: \text{ Two independent eigenvectors } v_3 = \begin{pmatrix} 1 \\ 0 \\ -1 \\ 1 \end{pmatrix}, v_4 = \begin{pmatrix} 0 \\ 1 \\ -2 \\ 2 \end{pmatrix} \text{ - If}$$

$$\alpha \neq 0: \text{ Find } v_3 \text{ eigenvector: } \begin{pmatrix} \alpha \\ -1 \\ -\alpha/2 \\ 1 \end{pmatrix} \text{ (up to scaling), and generalized eigenvector}$$

v_4 with $(A - I)v_4 = v_3$.

So:

$$P = \begin{cases} \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & -1 & -1 & -2 \\ 1 & 2 & 1 & 2 \end{pmatrix} & \alpha = 0 \\ \begin{pmatrix} 0 & 0 & \alpha & * \\ 0 & 0 & -1 & * \\ 1 & -1 & -\alpha/2 & * \\ 1 & 2 & 1 & * \end{pmatrix} & \alpha \neq 0 \end{cases}$$

where $*$ denotes the generalized eigenvector for $\lambda = 1$ when $\alpha \neq 0$.

Exercise 5:

1. Solve the differential system of 1st order :

$$\begin{cases} x_1'(t) = x_1(t) + x_2(t) \\ x_2'(t) = x_2(t) + x_4(t) \\ x_3'(t) = x_2(t) + x_3(t) \\ x_4'(t) = -x_1(t) + x_3(t) + 2x_4(t) \end{cases}$$

2. Solve the differential equation of the 3th order :

$$x_t^3 = -x_t^2 + x_t^1 + x_t + t$$

Correction

1. System: Write $X' = AX$ where $X = (x_1, x_2, x_3, x_4)^T$ and

$$A = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ -1 & 0 & 1 & 2 \end{pmatrix}$$

Characteristic polynomial: $\chi_A(\lambda) = (\lambda - 1)^4$ (compute to verify).

Check minimal polynomial: $A - I = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 1 & 1 \end{pmatrix}$ $(A - I)^2 \neq 0$, $(A - I)^3 = 0$,

so minimal polynomial $(\lambda - 1)^3$.

Jordan form: $J = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$ (blocks 3,1).

General solution: $X(t) = e^{tA}X_0 = Pe^{tJ}P^{-1}X_0$ where $e^{tJ} = \begin{pmatrix} e^t & te^t & \frac{t^2}{2}e^t & 0 \\ 0 & e^t & te^t & 0 \\ 0 & 0 & e^t & 0 \\ 0 & 0 & 0 & e^t \end{pmatrix}$.

Explicitly: Find P such that $A = PJP^{-1}$, then solution is $X(t) = Pe^{tJ}P^{-1}X(0)$.

2. Third order ODE: $x''' + x'' - x' - x = t$

Characteristic equation: $r^3 + r^2 - r - 1 = (r + 1)^2(r - 1) = 0$

Roots: $r = -1$ (double), $r = 1$

Homogeneous solution: $x_h(t) = (C_1 + C_2t)e^{-t} + C_3e^t$

Particular solution: Try $x_p(t) = At + B$

Then $x_p' = A$, $x_p'' = 0$, $x_p''' = 0$

Substitute: $0 + 0 - A - (At + B) = t \Rightarrow -At - (A + B) = t$

So $-A = 1 \Rightarrow A = -1$, and $-(A + B) = 0 \Rightarrow B = 1$

General solution: $x(t) = (C_1 + C_2t)e^{-t} + C_3e^t - t + 1$

Exercise 6:

Let

$$A = \begin{pmatrix} 2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 \end{pmatrix}$$

1. Give the minimal polynomial of A .
2. Give the shema of the basis of Jordan of A , deduce the dimension of $\text{Ker}(A - 2I_7)^r$ where $r \in \mathbb{N}^*$.

Correction

1. Minimal polynomial: Matrix A is already in Jordan form with eigenvalue $\lambda = 2$. The Jordan blocks are:

- Block 1: size 3 (rows 1-3)
- Block 2: size 1 (row 4)
- Block 3: size 1 (row 5)
- Block 4: size 2 (rows 6-7)

The largest Jordan block has size 3, so minimal polynomial is $(X - 2)^3$.

2. Jordan basis and kernel dimensions: The Jordan basis is already the standard basis $e_1, \dots, e_7$ with:

- Chain 1: $e_3 \rightarrow e_2 \rightarrow e_1$ (size 3)
- Chain 2: e_4 (size 1)
- Chain 3: e_5 (size 1)
- Chain 4: $e_7 \rightarrow e_6$ (size 2)

For $N = A - 2I_7$:

- $\dim(\ker N) = \text{number of Jordan blocks} = 4$
- $\dim(\ker N^2)$: From diagram, blocks of size ≥ 2 contribute: size3 $\rightarrow 2$, size2 $\rightarrow 2$, size1 $\rightarrow 1$ So: size3 block: 2 vectors, size2 block: 2 vectors, size1 blocks: 1 each $\rightarrow \text{total} = 2 + 2 + 1 + 1 = 6$
- $\dim(\ker N^3)$: All blocks contribute fully: size3 $\rightarrow 3$, size2 $\rightarrow 2$, size1 $\rightarrow 1 \rightarrow \text{total} = 3 + 2 + 1 + 1 = 7$
- For $r \geq 3$: $\dim(\ker N^r) = 7$

So:

$$\dim(\ker N) = 4, \quad \dim(\ker N^2) = 6, \quad \dim(\ker N^r) = 7 \text{ for } r \geq 3$$